

5 Animals use aerobic and anaerobic respiration to produce ATP.

(a) Aerobic respiration has three stages where reactions occur between enzymes and substrates.

(i) Which row of the table shows the location of each stage?

(1)

	Glycolysis	Link reaction	Krebs cycle
☐ A	cytoplasm	cytoplasm	cytoplasm
☐ B	cytoplasm	mitochondrial matrix	mitochondrial matrix
☐ C	mitochondrial matrix	cytoplasm	cytoplasm
☐ D	mitochondrial matrix	mitochondrial matrix	mitochondrial matrix

(ii) A student calculated the volume of oxygen used by a cockroach as $1.7 \text{ cm}^3 \text{ hour}^{-1}$. The same cockroach produced $1.3 \text{ cm}^3 \text{ hour}^{-1}$ of carbon dioxide in the same time.

What is the respiratory quotient (RQ) of this cockroach?

(1)

- ☐ **A** 0.7
- ☐ **B** 0.8
- ☐ **C** 1.3
- ☐ **D** 2.2

(iii) Hydramethylnon is an insecticide that is used to treat termite infestations.

Hydramethylnon inhibits the electron transport chain.

Which row of the table shows the expected changes caused by hydramethylnon?

(1)

	Oxygen consumption	Lactate production
☐ A	decrease	decrease
☐ B	decrease	increase
☐ C	increase	decrease
☐ D	increase	increase

(b) The table shows the oxygen consumption and body mass of two animals.

Animal	Oxygen consumption / $\text{cm}^3 \text{h}^{-1}$	Body mass /g
Southern elephant seal	1.82×10^4	5.00×10^6
Field mouse	20.00	10.00

These figures can be used to calculate the metabolic rate of each animal in $\text{cm}^3 \text{g}^{-1} \text{h}^{-1}$.

Calculate how many times greater the metabolic rate of the field mouse is compared with that of the southern elephant seal.

(2)

Answer

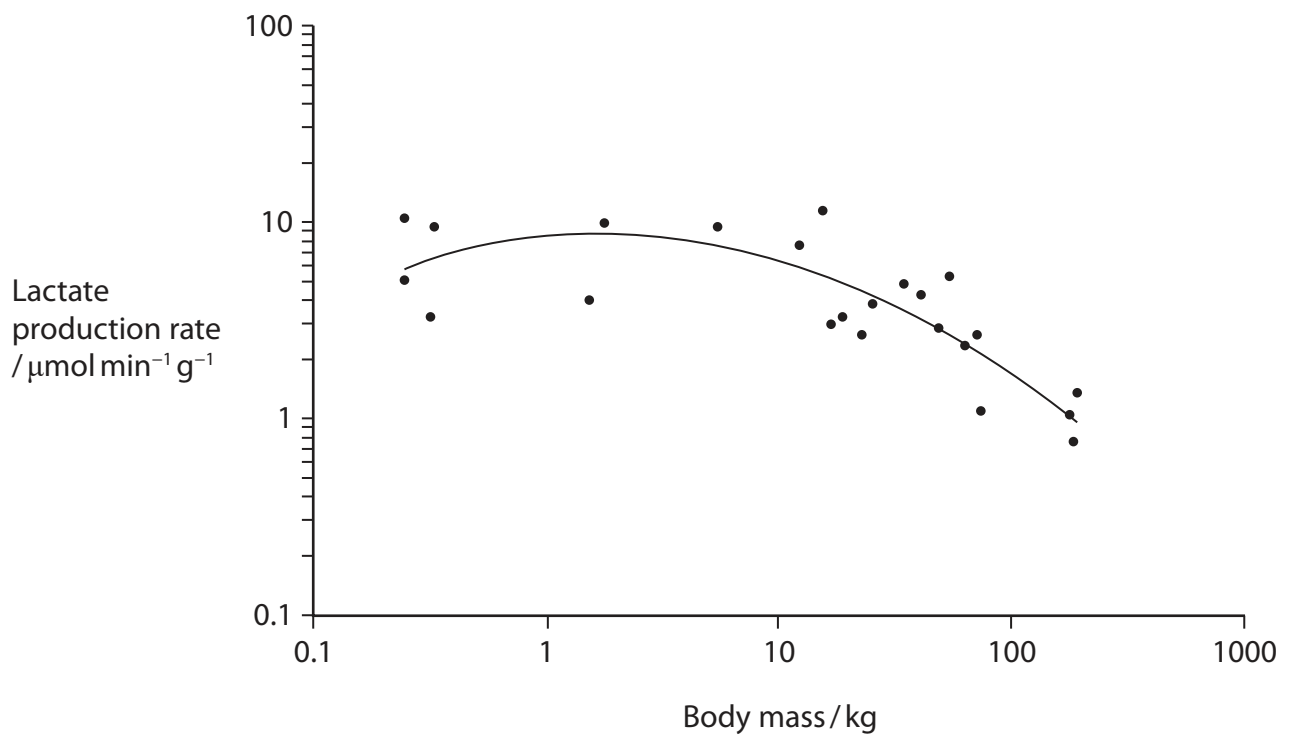
*(c) The photograph shows a crocodile (*Crocodylus porosus*) that lives in saltwater habitats.



(Source: © Danny Ye/Shutterstock)

An investigation measured the lactate production rate of these crocodiles when they were active.

The graph shows the results of this investigation.



Each plotted point on the graph represents a single crocodile.

Discuss the roles of the different types of respiration in these active crocodiles.

Use the information in the graph and your own knowledge to support your answer.

(6)

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(Total for Question 5 = 11 marks)

